AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM254
Managing IT: the why, the what and the how

Midterm Examination (MTA)
Fall Semester 2025/2026

Date: 20 /Nov./2025

Number of Exam Pages: 04 (including this cover sheet)	**Time Allowed:** 2 hours

Instructions:

- Total Marks: **60**
- This exam consists of **03** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

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PART 1: MCQ [10 Marks]

Each question carries a weight of 2 marks

Question 1: [10 marks] Read the following questions and choose the best answer.

1- What is the main goal of Knowledge Management in an organization?
   A. To ensure the right information is available at the right time and place for better decisions
   B. Impact To store as much data as possible
   C. To focus only on collecting raw data
   D. To replace human decision-making completely

2- Which type of team is made up of members who are separated by distance or time, even if they are in the same city?
   A. Project teams
   B. Virtual teams
   C. Operational teams
   D. Collocated team

3- Which of the following is not one of the five essential characteristics that distinguish services from physical goods?
   A. Strategic Perishability
   B. Tangibility
   C. Inseparability
   D. Non-ownership

4- Which of the following is not a component of the service portfolio?
   A. Service pipeline
   B. Service catalogue
   C. Retired services
   D. Service incidents

5- Which of the following is not one of the three characteristic properties that distinguish a database from a non-database system?
   A. Persistent
   B. Shared
   C. Temporary
   D. Interrelated

PART 2: Short questions [30 Marks]

Question 2: [6 marks]

Briefly describe the role of Service Operation and Service Transition within the ITIL framework.

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Question 3: [6 marks]

a) Describe the purpose of the Service Level Requirement (**SLR**) and the Service Level Agreement (**SLA**). (3 marks)

b) IT Service Continuity Management and Information Security Management are two key processes in Service Design. Describe the responsibilities of each. (3 marks)

Question 4: [5 marks]

a) Describe the Change Management process in ITIL Service Transition and specify the purpose of the Change Advisory Board (CAB). (3.5 marks)

b) What is the purpose of Release and Deployment Management, a process within ITIL Service Transition? (1.5 marks)

Question 5: [4 marks]

a) ITIL defines different 3 types of service providers. List 2 of these types and briefly describe each one. (2 marks)

b) Explain the difference between utility and warranty. (2 marks)

Question 6: [3 marks]

Match each step in the first column of the CSI approach with its correct description in the second column.

1-What is the vision	a- Ensure that improvements are sustained over time.
2-Where are we now?	b- Determine the target state for services or processes
3-Where do we want to be?	c- Measure and evaluate the results of improvements
4-How do we get there?	d- Define the overall goals and objectives for the service or the improvement initiative
5-Did we get there?	e- Plan the actions required to reach the target state
6-How do we keep the momentum going?	f- Assess the current state of services, processes, and performance metrics

Question 7: [3 marks]

What are the three dimensions of requirements engineering? Provide a brief description of each.

Question 8: [3 marks]

Explain the purpose of the Analysis, Design, and Testing stages in the Waterfall software development model.

Part 3: Problem Solving Questions [20 Marks]

Question 9: [20 marks]

StreamFlix, a global streaming service provider, offers customers access to movies, TV shows, and live events through its online platform. The service ensures secure

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handling of user accounts and payment information. A support center operates daily to handle tasks such as account registration, password resets, and assisting with software-related issues on both web and mobile applications.

The average monthly uptime is 99.8%, the average incident resolution time is 1 hour, and the customer satisfaction score is 4.5/5. These values allowed StreamFlix to compare current performance against expected standards and identify areas for improvement.

StreamFlix's vision is to have happy customers and sustainable revenue. Therefore, maintaining high service availability, fast content delivery, and minimal downtime is essential and should be regularly reviewed to ensure alignment with business goals.

On a Friday evening, users began reporting that they were unable to stream content, and playback errors were occurring across multiple devices. By 7:15 PM, the service desk classified the issue as a critical incident due to its widespread impact on streaming operations and user experience and detected a sudden disruption in the online streaming service.

Initial checks showed no issues with the streaming application itself, so the incident was escalated to the IT infrastructure team. Within 40 minutes, they discovered that a recent content delivery network (CDN) configuration update had inadvertently blocked access to several media servers. The configuration was corrected immediately, and streaming services were fully restored by 8:10 PM.

After resolving the incident, the engineering team conducted a root cause analysis, which revealed that the CDN update had not been tested in a staging environment before deployment. To prevent future disruptions, StreamFlix implemented stricter change management procedures and mandatory testing for all CDN and server updates. These preventive measures were documented in the problem management system and communicated to the IT and operations teams.

Average uptime, incident resolution time, and customer satisfaction scores, which act as indicators, were monitored to show whether the CSFs are being met. Analysis showed that the mean time to resolve this incident met the internal targets.

a) What steps were taken in incident management to restore services? (3 marks)

b) How did problem management address the root cause of the outage? (3 marks)

c) Identify and define the Incident, Event, and Problem in this scenario. (4 marks)

d) Identify all the stakeholders involved in this service. (3 marks)

e) What are the functional, and compliance requirement of the streaming service? (3 marks)

f) Based on the scenario, list the baseline, the Critical Success Factors (CSF) and the corresponding Key Performance Indicators (KPI) for StreamFlix. (4 marks)

End of questions

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